

All Dice, No Slice: Metric Artifacts, Data Leakage, and Task Interference in Multi-Task Computational Pathology

Muhammad Abdullah Ali^{a,*}, Muhammad Ibrahim Kiani^a, Muhammad Abdullah Aamir^a

^a*Department of Data Science, National University of Computer and Emerging Sciences (FAST-NUCES), Islamabad, Pakistan*

Abstract

Simultaneous classification and lesion segmentation via hard parameter-sharing U-Nets is an appealing paradigm for computational pathology. Recently, classification accuracies of 82%–90% and Dice scores of 97%–99% were reported across diverse medical imaging modalities using standard convolutional backbones and static loss weighting. We conduct a 26-experiment replication and methodological audit across four datasets: TCGA-LGG brain-tumor MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and the 19-tissue PanNuke benchmark. Under strict patient-level data isolation (**GroupKFold**), the claimed accuracies are not reproducible, and the reported Dice levels are attainable only as empty-mask artifacts: all four of our SIIM configurations converge to all-empty predictions yet report 77.74% Dice — exactly the lesion-free fraction of the validation split under the convention that scores empty-over-empty slices as perfect agreement. On fine-grained 6-class prostate cancer grading (PANDA), empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34% across all 12 evaluated configurations, with static 5:1-weighted 10^{-3} baselines (Runs 03–04) reaching 45.15% – 46.15% / 43.30% – 44.34% (vs. 88.0% and 99.0% claimed). Three systemic causes explain the discrepancy: (1) empty-mask Dice crediting, (2) patient-level data leakage, and (3) task interference from the joint optimiza-

*Corresponding author

Email addresses: i232523@isb.nu.edu.pk (Muhammad Abdullah Ali), i232536@isb.nu.edu.pk (Muhammad Ibrahim Kiani), i232538@isb.nu.edu.pk (Muhammad Abdullah Aamir)

tion recipe. Matched ablations further show that skip connections are a null result, whereas removing Macenko stain normalization *improves* fine-grained classification (+5.51% on PANDA, +2.68% on PanNuke). Our benchmark and audit protocol are released to prevent such metric artifacts in future multi-task pathology studies.

Keywords: Computational Pathology, Multi-Task Learning, U-Net, Reproducibility, Metric Artifacts, Data Leakage, GradNorm

1. Introduction

Deep learning systems in digital pathology and radiology are increasingly required to deliver both diagnostic categorization and lesion delineation [1, 2]. Large visual foundation models (e.g., UNI [3], CONCH [4], and Virchow [5]) show that clinically actionable representations require massive pretraining and self-supervised objectives across gigapixel whole-slide images (WSIs). In resource-constrained deployments, hard parameter sharing via an encoder–decoder backbone (a U-Net [6] with dual prediction heads) remains a lightweight alternative that couples both objectives in a single forward pass.

Recently, Rhanoui et al. [7] claimed that a standard U-Net with lightweight encoders (VGG16 [8] and MobileNetV2 [9]) trained with static loss weights ($\lambda_{seg} = 5.0, \lambda_{cls} = 1.0$) and a high learning rate ($\eta = 10^{-3}$) simultaneously solves both classification and segmentation across four distinct medical modalities (Brain MRI, Skin Dermoscopy, Prostate Histopathology, and Chest X-Ray), reporting 82% – 90% classification accuracy and 97% – 99% Dice similarity coefficients.

In this paper, we report a 26-run replication study and systematic methodological audit. Our findings demonstrate that:

1. **Empirical Irreproducibility:** When evaluated with strict patient/biopsy-level isolation (GroupKFold), multi-class grading on PANDA prostate biopsies collapses from the claimed 88.0% accuracy / 99.0% Dice down to an empirical range of 29.04% – 46.15% Acc / 28.94% – 44.34% Dice across all 12 configurations, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving 45.15% – 46.15% / 43.30% – 44.34%.
2. **The Empty-Mask Dice Metric Artifact:** Near-perfect (99%) Dice claims on sparse-lesion datasets such as SIIM-ACR pneumothorax can arise from crediting lesion-free slices ($|Y| = |\hat{Y}| = 0$) as Dice = 1.0; our

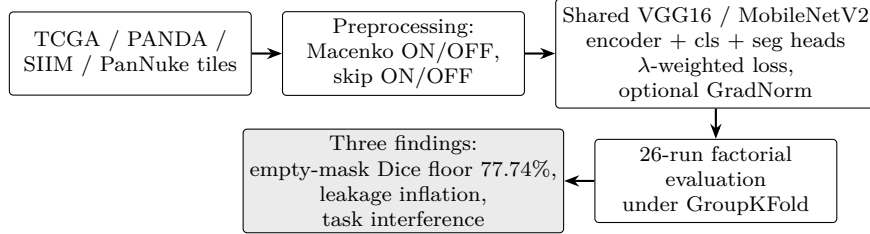


Figure 1: Overview of the 26-run replication and audit pipeline. Four source datasets (TCGA-LGG, PANDA, SIIM-ACR, PanNuke) are tiled and passed through a preprocessing stage that toggles Macenko stain normalization and skip connections. A shared VGG16/MobileNetV2 encoder feeds a classification head and a segmentation head trained with a λ -weighted loss (static ratios or optional GRADNORM). All 26 configurations are evaluated under patient-level GroupKFold isolation, yielding three findings: the empty-mask Dice floor of 77.74% on SIIM, patient-level leakage inflation of classification accuracy, and task interference from the joint optimization recipe.

four SIIM runs measure exactly this floor (77.74%) under an all-empty predictor.

3. **Controlled Multi-Task & Ablation Analysis:** Through strictly matched factorial controls (learning-rate pairs, static loss-weight ratios $\lambda_{seg} : \lambda_{cls} \in \{1:10, 1:1, 5:1, 10:1\}$, GRADNORM [10] dynamic balancing, Macenko normalization, and skip-connection ablations), we isolate how individual components govern multi-task optimization: dynamic gradient balancing actively impairs fine-grained classification (45.39% $\rightarrow$ 29.04% Acc), while skip connections are a verified null result.

To facilitate complete reproducibility, all training scripts, raw JSON summaries for all 26 runs, evaluation checkpoints, and generated figures are publicly available at <https://github.com/ApatheticMioz/cancer-pathology-dl> [11]. Figure 1 summarizes the end-to-end pipeline and the three findings it yields.

2. Multi-Task Formulation & Evaluation Metrics

2.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ denote an input image tile. A shared convolutional encoder $f_\theta(\mathbf{x})$ extracts a bottleneck latent representation $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$. The classification branch applies global average pooling (GAP) followed by a two-layer multilayer perceptron (hidden width 256, dropout 0.5):

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}(\mathbf{W}_2 \phi(\mathbf{W}_1 \text{GAP}(\mathbf{z}) + \mathbf{b}_1) + \mathbf{b}_2) \in \Delta^{K-1}, \quad (1)$$

where K is the number of diagnostic classes, Δ^{K-1} denotes the probability simplex, and $\phi(\cdot)$ is the ReLU nonlinearity. The segmentation branch passes $\mathbf{z}$ through a decoder Dec_ϕ with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^L$ from encoder stages, yielding unnormalized pixel logits $\mathbf{l}_{seg} \in \mathbb{R}^{H \times W \times M}$. For binary segmentation ($M = 1$), $\hat{\mathbf{y}}_{seg} = \sigma(\mathbf{l}_{seg})$ with $\sigma(\cdot)$ the sigmoid; for $M > 1$, class probabilities are obtained via channel-wise Softmax.

The joint multi-task training objective is defined as:

$$\mathcal{L}_{total}(\theta, \phi, \mathbf{W}_{cls}) = \lambda_{cls} \mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg} \mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy $\mathcal{L}_{CE}$. In the published study of Rhanoui et al. [7], the segmentation loss was stated as $\mathcal{L}_{seg} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}$; the implementation audited for this study, however, optimizes purely cross-entropy objectives, without any Dice-based penalty term (a documented deviation):

$$\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}) = \begin{cases} \mathcal{L}_{BCE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M = 1 \text{ (binary tasks)}, \\ \mathcal{L}_{CE}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), & M > 1 \text{ (multi-class tasks)}. \end{cases} \quad (3)$$

The Dice similarity coefficient (DSC) is evaluated post-hoc as a validation metric, not as an active loss term during gradient descent.

2.2. The Empty-Mask Dice Metric Artifact

The Sørensen–Dice coefficient between ground-truth mask Y and binary prediction $\hat{Y}$ ($Y, \hat{Y} \in \{0, 1\}^{H \times W}$) is:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad \epsilon = 10^{-8}; \quad \text{Dice} := 1 \text{ if } |Y| + |\hat{Y}| = 0. \quad (4)$$

Empty references and predictions are recognized validation pitfalls in biomedical image analysis [12]; two conventions govern empty slices:

- **Foreground-Only Evaluation:** Compute Dice strictly over non-empty target slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** Assign $\text{Dice} = 1.0$ when $|Y| = 0 \wedge |\hat{Y}| = 0$, and $\text{Dice} = 0.0$ when $|Y| = 0 \wedge |\hat{Y}| > 0$.

Averaged over all slices under empty-credit assignment, the macroscopic mean is

$$\overline{\text{Dice}}_{\text{all}} = \frac{1}{N} \sum_{i=1}^N d_i, \quad d_i = \text{Dice}(Y_i, \hat{Y}_i) \text{ if } |Y_i| > 0, \text{ else } \mathbf{1}[|\hat{Y}_i| = 0], \quad (5)$$

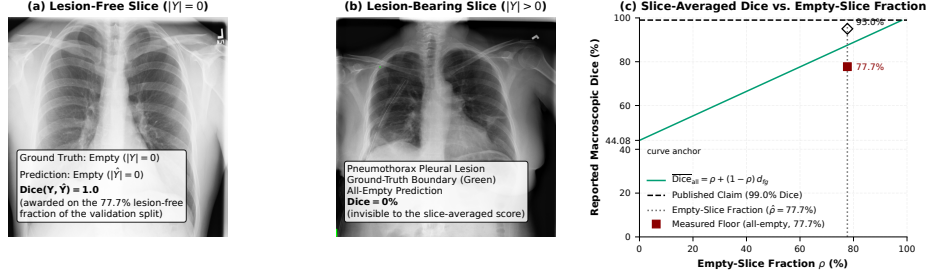


Figure 2: The empty-mask Dice artifact on SIIM-ACR pneumothorax. (a) Lesion-free slice ($|Y| = 0$): under empty-credit assignment an all-empty prediction is awarded Dice = 1.0; lesion-free slices constitute $\hat{\rho} = 77.7\%$ of our validation split. (b) Lesion-bearing slice ($|Y| > 0$, ground-truth boundary in green): our collapsed models predict empty masks here and score Dice = 0, invisible to the slice-averaged mean. (c) Convex-combination curve (Eq. 6): a foreground Dice of 77.74% would report 95.04% at $\hat{\rho} = 0.777$ (analytic illustration), whereas all four of our SIIM runs measure the all-empty floor of 77.74% itself.

which for mean foreground Dice $\overline{\text{Dice}}_{\text{fg}}$ on lesion-bearing slices is the convex combination

$$\overline{\text{Dice}}_{\text{all}} = \rho + (1 - \rho) \cdot \overline{\text{Dice}}_{\text{fg}}, \quad (6)$$

where ρ is the proportion of lesion-free slices. In sparse pathologies such as SIIM-ACR pneumothorax, a model with a genuine foreground Dice of 77.74% would, as an analytic illustration, report $\overline{\text{Dice}}_{\text{all}} = 0.777 \times 1.0 + 0.223 \times 0.7774 = 95.04\%$ at the observed lesion-free fraction $\hat{\rho} = 1,659/2,135 = 0.777$. Our four SIIM runs realize the more extreme failure mode: emitting empty masks on every validation slice throughout training, each reported 77.74% (Runs 05, 06, 11, 12) *is the all-empty floor itself* — Eq. 5 with zero foreground Dice. As Fig. 2 shows, slice-averaged Dice on sparse data reports the lesion-free fraction rather than lesion overlap, so published 99.0% Dice claims [7] are uninterpretable as evidence of lesion localization.

2.3. Adaptive Gradient Balancing via GradNorm

To evaluate whether dynamic loss balancing mitigates cross-task gradient competition, we introduced a parameterized variant of GRADNORM [10] in Phase v2.

Let W denote the shared convolutional filters and normalization layers of the U-Net encoder f_{θ} . The L_2 gradient norm of task $i \in \{\text{seg}, \text{cls}\}$ with respect to W at minibatch t is:

$$G_W^{(i)}(t) = \|\nabla_W w_i(t) \mathcal{L}_i(t)\|_2. \quad (7)$$

For stability in the primary Adam optimizer, balancing weights are learnable log-weights $\mu_i(t) = \ln w_i(t)$ (initialized to $w_{seg}(0) = 5.0$, $w_{cls}(0) = 1.0$), updated jointly with the network parameters. Minibatch optimization minimizes the L_1 gradient disparity objective:

$$\mathcal{L}_{grad}(t) = \sum_{i \in \{seg, cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t) \cdot [r_i(t)]^\alpha \right|, \quad (8)$$

where $\bar{G}_W(t) = \frac{1}{2} \sum_j G_W^{(j)}(t)$ is the average gradient norm across tasks, $\alpha = 1.5$ is the asymmetry parameter, and $r_i(t)$ represents the relative inverse training rate:

$$r_i(t) = \frac{\tilde{\mathcal{L}}_i(t)}{\frac{1}{2} \sum_{j \in \{seg, cls\}} \tilde{\mathcal{L}}_j(t)}, \quad \text{with} \quad \tilde{\mathcal{L}}_i(t) = \frac{\mathcal{L}_i(t)}{\mathcal{L}_i(0)}, \quad (9)$$

where $\mathcal{L}_i(0)$ is the initial step-0 loss. Following Chen et al. [10], slower-training tasks exhibit higher loss ratios $r_i(t) > 1$, raising their target gradient norm $\bar{G}_W(t)[r_i(t)]^\alpha$. After each optimizer update, weights are strictly renormalized such that $\sum_i w_i(t) = 2$. In the canonical formulation [10], $\mathcal{L}_{grad}$ is differentiated against detached targets and applied to the balancing weights alone through a dedicated optimizer (learning rate 0.025); §4.3 probes this distinction empirically. Evaluating this module across Groups 2 and 4 isolates how dynamic gradient equalization behaves in hard-parameter-sharing architectures.

2.4. Color Deconvolution and Optical Density Space

To mitigate staining variability across clinical laboratories, the v2 pipeline incorporates Macenko stain normalization [13]. RGB pixel intensities $\mathbf{I} \in [0, 255]^3$ are converted to optical density (OD) space:

$$\mathbf{OD} = -\ln \left(\frac{\mathbf{I} + \epsilon}{I_0} \right), \quad (10)$$

where $I_0 = 255$ represents transmitted light intensity. The underlying implementation computes optical density via natural logarithms $-\ln(\cdot)$ rather than the canonical base-10 formulation $-\log_{10}(\cdot)$ [13]. Hematoxylin and eosin stain vectors are estimated by eigendecomposition of the optical-density covariance matrix (stain-vector angles restricted to the 1st–99th percentile range) over OD values above a threshold of 0.15, and projected onto a target reference vector. While this effectively eliminates macroscopic batch

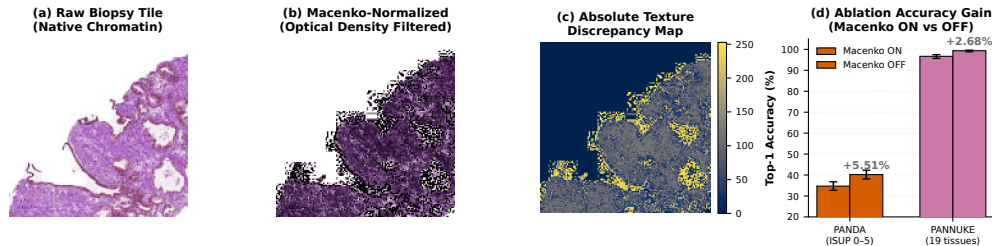


Figure 3: Visual and quantitative impact of Macenko stain normalization on fine-grained computational pathology. (a) Raw prostate biopsy tile. (b) Optical-density-projected tile: color deconvolution acts as a chrominance low-pass filter, flattening sub-cellular texture. (c) Absolute texture discrepancy map. (d) Empirical top-1 accuracy gains across matched models when Macenko normalization is bypassed (+5.51% on PANDA, +2.68% on PanNuke).

variations, projecting raw RGB channels onto static stain vectors acts as a chrominance low-pass filter, smoothing sub-cellular chromatin textures that are clinically essential for grading fine glandular architectures (visualized in Fig. 3).

3. The 26-Run Empirical Reproduction Campaign

3.1. Datasets and Experimental Protocol

We evaluated four benchmark datasets:

1. **TCGA-LGG Brain Tumor:** 2-class lesion classification and FLAIR abnormality segmentation on 2D 256×256 slices ($N = 3,929$; 110 patients [14]).
2. **PANDA Prostate Biopsies:** 6-class ISUP Gleason grading and gland segmentation ($N = 10,616$ tiles [15]; patient-level **GroupKFold** isolation).
3. **SIIM-ACR Pneumothorax:** Binary chest-radiography detection and pleural boundary segmentation ($N = 12,047$ images [16]).
4. **PanNuke:** 19-tissue multi-organ benchmark ($N = 7,901$ patches [17]); having no published multi-task targets, it serves as an unclaimed external control.

We trained all models with PyTorch (up to 50 epochs with early stopping, patience 10; batch size 32; fixed seed 42) using ImageNet-pretrained encoders and the Adam optimizer with a constant per-phase learning rate

and zero weight decay, and we applied no learning-rate schedule or additional regularization. In Phase v1 (Runs 01–06), we replicated the original static setup ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw inputs). In Phase v2 (Runs 07–12), we incorporated dynamic GRADNORM ($\alpha = 1.5$, $\eta = 10^{-4}$) and Macenko stain normalization [13]. Groups 3–5 (Runs 13–26) systematically evaluate multi-organ generalizability, optimization hyperparameter isolation, and architectural skip ablations. Table 1 summarizes all 26 runs and Figure 4 visualizes them.

4. Discussion & Methodological Anatomy

4.1. The Anatomy of the PANDA Collapse

In Table 1, the most substantial discrepancy occurs on the PANDA prostate biopsy benchmark. The original authors reported 88.0% accuracy on 6-class ISUP grading alongside a 99.0% Dice score. In contrast, across all 12 PANDA configurations tested, empirical top-1 accuracy spans 29.04% – 46.15% and Dice spans 28.94% – 44.34%, with the static 5:1-weighted 10^{-3} baselines (Runs 03–04) achieving 45.15% / 43.30% (VGG16) and 46.15% / 44.34% (MobileNetV2); Run 04 is the strongest single configuration in our campaign.

This outcome is consistent with established clinical and machine learning benchmarks. In the international PANDA challenge (1,010 participating teams) [15], the top-performing whole-slide multi-instance learning algorithms reached external validation quadratic weighted κ of 0.862 (95% CI 0.840–0.884) and 0.868 (0.835–0.900) on U.S. and European cohorts, and inter-observer Gleason grading of biopsy cores is itself imperfect ($\kappa = 0.68$ among urologic pathologists and 0.435 among general pathologists [18, 19]). Claiming 88.0% accuracy on 128×128 isolated patches with a standard 2D CNN is infeasible without patient-level data leakage (tiles from the same biopsy core contaminating both train and test splits). Figure 5 traces the PANDA collapse across training epochs and isolates the responsible mechanism.

4.2. Controlled Skip Connection Ablation: A Verified Null Result

Skip connections are evaluated in strictly matched pairs (learning rate, GradNorm, Macenko normalization held identical):

- **Coarse macro-tumor structures (TCGA):** Run 25 (skip-ablated) vs. matched baseline Run 08 (both $\eta = 10^{-4}$, GradNorm, Macenko): accuracy 93.32% $\rightarrow$ 94.34% ($\Delta_{\text{Acc}} = +1.02$), Dice 84.20% vs 84.12% ($\Delta_{\text{Dice}} = -0.08$).
- **Fine micro-glandular structures (PANDA):** Run 10 (34.70% Acc / 31.34% Dice) vs. matched skip-ablated Run 26 (35.31% / 28.94%): $\Delta_{\text{Acc}} = +0.61$, $\Delta_{\text{Dice}} = -2.40$.

In both cases the differences are small, and the 95% Wilson score intervals on top-1 accuracy overlap substantially (TCGA: [91.34, 94.87] vs [92.49, 95.76]; PANDA: [32.69, 36.76] vs [33.30, 37.38]). We therefore report these as *verified null results*: the bottleneck-only representation suffices for both coarse macro-lesion and fine micro-glandular segmentation; zeroing the skips produces no detectable penalty in a single-run comparison. A definitive directional claim would require multi-seed replication of the matched pairs.

Macenko stain-normalization ablation.. The matched Macenko pair (Group 5, Runs 23–24) is the only single-run ablation with a statistically resolvable classification shift: omitting normalization raises PANDA top-1 accuracy by +5.51 and PanNuke accuracy by +2.68 (disjoint 95% CIs), while Dice stays flat on PANDA and rises by +12.79 on PanNuke. Color deconvolution thus harms fine-grained classification, consistent with over-smoothing of the chromatin/texture cues on which grading relies, and partially decomposes the v1-vs-v2 PANDA gap of §4.1. Figure 6 consolidates all six matched pairs; Figure 3 shows the texture flattening.

4.3. Isolating Learning Rate from Dynamic Gradient Balancing

Group 4 (PANDA $\times$ VGG16) factorizes the key optimization hyperparameters: learning rate ($\eta \in \{10^{-3}, 10^{-4}\}$), dynamic gradient balancing (GradNorm on/off), and static loss-weighting ratios $\lambda_{\text{seg}} : \lambda_{\text{cls}} \in \{1:10, 1:1, 5:1, 10:1\}$, all under raw inputs.

Direct Isolation of Dynamic Gradient Balancing (Run 18).. The isolated GradNorm arm at $\eta = 10^{-3}$ under strictly matched raw-input conditions (Run 18: PANDA $\times$ VGG16, GradNorm $\alpha = 1.5$, raw input) compares directly against the static 5:1 baseline (Run 20). Activating GradNorm ($\alpha = 1.5$) causes validation accuracy to collapse from 45.39% down to 29.04% (95% Wilson CI: [27.14, 31.02]), alongside a corresponding collapse in segmentation

Dice from 44.08% down to 31.43% ($\Delta_{\text{Acc}} = -16.35\%$, $\Delta_{\text{Dice}} = -12.65\%$, with non-overlapping confidence intervals: $[43.27, 47.52]$ vs $[27.14, 31.02]$).

This 29.04% accuracy and 31.43% Dice matches the performance observed in Phase v2 at $\eta = 10^{-4}$ (Run 09: 30.89% Acc / 31.59% Dice). The $\approx 30\%$ floor on fine-grained PANDA grading is thus driven primarily by dynamic gradient balancing ($\alpha = 1.5$), whose gradient-norm equalization forces destructive updates onto the shared encoder, rather than by the lower learning rate.

Learning Rate and Loss Weighting Dynamics.. Holding GradNorm off, reducing the learning rate from $\eta = 10^{-3}$ (Run 20: 45.39% Acc) to $\eta = 10^{-4}$ (Run 17: 43.35%) yields a minor change within overlapping 95% CIs. The static loss-ratio sweep (Runs 19–22) confirms $\lambda = 5:1$ as the best static balance, yet no ratio approaches the claimed 88% (Figure 7). The isolated GradNorm arm (Run 18) collapses to 29.04% against the 45.39% static baseline with non-overlapping CIs, and the canonical decoupled scheme plateaus at 43.06% — dynamic balancing in either formulation cannot bridge the gap (Figures 5, 6).

5. Conclusion & Recommendations

Across 26 experiments, the claimed 88% – 99% metrics did not reproduce under patient isolation. The residual gap stems from dataset difficulty and the joint optimization recipe: skip connections are a null result, while dynamic gradient balancing ($\alpha = 1.5$) destabilizes classification (45.39% $\rightarrow$ 29.04% Acc). We recommend: (1) mandatory patient/biopsy-level **GroupKFold** splitting; (2) explicit reporting of the Dice evaluation convention, with foreground-only or zero-penalty scoring on sparse-lesion datasets; (3) 95% Wilson score intervals on matched single-run ablation comparisons to distinguish genuine architectural effects from stochastic variance; (4) validation on diverse multi-organ benchmarks such as PanNuke; and (5) systematic evaluation of multi-task balancing hyperparameters before clinical translation.

References

- [1] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. Van Der Laak, B. Van Ginneken, C. I. Sánchez, A survey on deep learning in medical image analysis, *Medical Image Analysis* 42 (2017) 60–88.

- [2] G. Campanella, M. G. Hanna, L. Geneslaw, A. Miraflor, V. Werneck Krauss Silva, F. Klauschen, et al., Clinical-grade computational pathology using weakly supervised deep learning on whole slide images, *Nature Medicine* 25 (8) (2019) 1301–1309.
- [3] R. J. Chen, T. Ding, M. Y. Lu, D. F. Williamson, G. Jaume, B. Chen, et al., Towards a general-purpose foundation model for computational pathology, *Nature Medicine* 30 (3) (2024) 850–862. [doi:10.1038/s41591-024-02857-3](https://doi.org/10.1038/s41591-024-02857-3).
- [4] M. Y. Lu, B. Chen, D. F. Williamson, M. Zhao, A. K. Chow, K. Ike-mura, et al., A visual-language foundation model for computational pathology, *Nature Medicine* 30 (3) (2024) 863–874. [doi:10.1038/s41591-024-02856-4](https://doi.org/10.1038/s41591-024-02856-4).
- [5] E. Vorontsov, A. Bozkurt, A. Casson, G. Shaikovski, M. Zelechowski, K. Severson, et al., A foundation model for clinical-grade computational pathology and rare cancers detection, *Nature Medicine* 30 (10) (2024) 2924–2935. [doi:10.1038/s41591-024-03141-0](https://doi.org/10.1038/s41591-024-03141-0).
- [6] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Springer, 2015, pp. 234–241.
- [7] M. Rhanoui, K. Alaoui Belghiti, M. Mikram, Multi-task deep learning for simultaneous classification and segmentation of cancer pathologies in diverse medical imaging modalities, *Onco* 5 (3) (2025) 34. [doi:10.3390/onco5030034](https://doi.org/10.3390/onco5030034).
- [8] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, *arXiv preprint arXiv:1409.1556* (2014).
- [9] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, in: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510–4520.
- [10] Z. Chen, V. Badrinarayanan, C.-Y. Lee, A. Rabinovich, Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask

- networks, in: International Conference on Machine Learning (ICML), PMLR, 2018, pp. 794–803.
- [11] M. A. Ali, M. I. Kiani, M. A. Aamir, Multi-task deep learning in cancer pathology: Replication and empirical audit, <https://github.com/ApatheticMioz/cancer-pathology-dl>, GitHub Repository (2026).
 - [12] A. Reinke, M. D. Tizabi, M. Baumgartner, M. Eisenmann, et al., Understanding metric-related pitfalls in image analysis validation, *Nature Methods* 21 (2) (2024) 182–194. doi:10.1038/s41592-023-02150-0.
 - [13] M. Macenko, M. Niethammer, J. Marron, D. Borland, J. T. Woosley, X. Guan, C. Schmitt, N. E. Thomas, A method for normalizing histology slides for quantitative analysis, in: IEEE International Symposium on Biomedical Imaging (ISBI), IEEE, 2009, pp. 1107–1110. doi:10.1109/ISBI.2009.5193250.
 - [14] M. Buda, A. Saha, M. A. Mazurowski, Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm, *Computers in Biology and Medicine* 109 (2019) 218–225. doi:10.1016/j.compbiomed.2019.05.002.
 - [15] W. Bulten, K. Kartasalo, P.-H. C. Chen, B. Delahunt, H. Pinckaers, K. Nagpal, M. Valkonen, G. Litjens, et al., Artificial intelligence for diagnosis and gleason grading of prostate cancer: the panda challenge, *Nature Medicine* 28 (1) (2022) 154–163. doi:10.1038/s41591-021-01620-2.
 - [16] Society for Imaging Informatics in Medicine (SIIM) and American College of Radiology (ACR), SIIM-ACR pneumothorax segmentation, <https://kaggle.com/competitions/siim-acr-pneumothorax-segmentation>, kaggle competition dataset (2019).
 - [17] J. Gamper, N. Alemi Koohbanani, K. Benet, A. Khuram, N. Rajpoot, Pannuke: an open pan-cancer histology dataset for nuclei instance segmentation and classification, *Computers in Medical Imaging and Graphics* 80 (2020) 101703. doi:10.1016/j.compmedimag.2019.101703.
 - [18] W. C. Allsbrook, K. A. Mangold, M. H. Johnson, R. B. Lane, C. G. Lane, J. I. Epstein, Interobserver reproducibility of Gleason grading

of prostatic carcinoma: urologic pathologists, Human Pathology 32 (1) (2001) 74–80. [doi:10.1053/hupa.2001.21134](https://doi.org/10.1053/hupa.2001.21134).

- [19] W. C. Allsbrook, K. A. Mangold, M. H. Johnson, R. B. Lane, C. G. Lane, J. I. Epstein, Interobserver reproducibility of Gleason grading of prostatic carcinoma: general pathologist, Human Pathology 32 (1) (2001) 81–88. [doi:10.1053/hupa.2001.21135](https://doi.org/10.1053/hupa.2001.21135).

Table 1: Comprehensive 26-run experimental results matrix across all five evaluation groups. V1 = original static baseline ($\eta = 10^{-3}$, $\lambda_{seg} : \lambda_{cls} = 5 : 1$, raw input); V2 = GradNorm variant ($\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization). $\checkmark$ = GradNorm active; $-$ = not used. Markers: $\times$ = skip connections zeroed; $\diamond$ = raw input without Macenko normalization (v1 runs are raw by construction). Rows 17–22 (Group 4) systematically isolate learning rate ($\eta = 10^{-4}$ vs 10^{-3}), GradNorm ($\alpha = 1.5$, Run 18), and loss weighting ratios on PANDA $\times$ VGG16 under raw input. Pub. Acc/Dice: published reference targets of Rhanoui et al. [7] ($-$ = no published target). Δ_{Acc} / Δ_{Dice} : difference relative to published targets. All metrics in %. Complete 95% Wilson score intervals for all 26 configurations are tabulated in the open-source repository matrix (`paper_results_matrix_with_ci.csv`).

#	Configuration	Dataset	Encoder	LR	η	GN	Val	Acc	Val	Dice	Pub.	Acc	Pub.	Dice	Δ_{Acc}	Δ_{Dice}
Group 1: Baseline Reproductions (static 5:1, $\eta = 10^{-3}$, raw input)																
01	Naked Baseline	TCGA	VGG16	10^{-3}	—	—	88.82	76.02	89.0	97.0	—	97.0	—	—	−0.18	−20.98
02	Naked Baseline	TCGA	MobileNetV2	10^{-3}	—	—	94.60	87.56	90.0	98.0	—	98.0	—	—	+4.60	−10.44
03	Naked Baseline	PANDA	VGG16	10^{-3}	—	—	45.15	43.30	87.0	98.0	—	98.0	—	—	−41.85	−54.70
04	Naked Baseline	PANDA	MobileNetV2	10^{-3}	—	—	46.15	44.34	88.0	99.0	—	99.0	—	—	−41.85	−54.66
05	Naked Baseline	SIIM	VGG16	10^{-3}	—	—	83.19	77.74	82.0	99.0	—	99.0	—	—	+1.19	−21.26
06	Naked Baseline	SIIM	MobileNetV2	10^{-3}	—	—	83.47	77.74	87.0	99.0	—	99.0	—	—	−3.53	−21.26
Group 2: V2 Package (GradNorm $\alpha = 1.5$, $\eta = 10^{-4}$, Macenko normalization)																
07	GradNorm v2	TCGA	VGG16	10^{-4}	$\checkmark$	$\checkmark$	92.80	78.30	89.0	97.0	—	97.0	—	—	+3.80	−18.70
08	GradNorm v2	TCGA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	93.32	84.20	90.0	98.0	—	98.0	—	—	+3.32	−13.80
09	GradNorm v2	PANDA	VGG16	10^{-4}	$\checkmark$	$\checkmark$	30.89	31.59	87.0	98.0	—	98.0	—	—	−56.11	−66.41
10	GradNorm v2	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	34.70	31.34	88.0	99.0	—	99.0	—	—	−53.30	−67.66
11	GradNorm v2	SIIM	VGG16	10^{-4}	$\checkmark$	$\checkmark$	81.08	77.74	82.0	99.0	—	99.0	—	—	−0.92	−21.26
12	GradNorm v2	SIIM	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	83.00	77.74	87.0	99.0	—	99.0	—	—	−4.00	−21.26
Group 3: PanNuke 19-Tissue Multi-Organ Benchmark (external control; no published targets)																
13	Naked Baseline	PanNuke	VGG16	10^{-3}	—	—	98.85	72.48	—	—	—	—	—	—	—	—
14	Naked Baseline	PanNuke	MobileNetV2	10^{-3}	—	—	99.43	74.21	—	—	—	—	—	—	—	—
15	GradNorm v2	PanNuke	VGG16	10^{-4}	$\checkmark$	$\checkmark$	91.90	39.59	—	—	—	—	—	—	—	—
16	GradNorm v2	PanNuke	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	96.68	60.54	—	—	—	—	—	—	—	—
Group 4: Optimization Isolation (PANDA $\times$ VGG16: LR, GradNorm, loss weighting; raw input)																
17	Isolate LR: static $\eta = 10^{-4}$ $\diamond$	PANDA	VGG16	10^{-4}	—	—	43.35	37.99	87.0	98.0	—	98.0	—	—	−43.65	−60.01
18	Isolate GradNorm: $\alpha = 1.5$	PANDA	VGG16	10^{-3}	$\checkmark$	$\checkmark$	29.04	31.43	87.0	98.0	—	98.0	—	—	−57.96	−66.57
19	Loss weight $\lambda = 1:1$	PANDA	VGG16	10^{-3}	—	—	42.73	41.72	87.0	98.0	—	98.0	—	—	−44.27	−56.28
20	Loss weight $\lambda = 5:1$	PANDA	VGG16	10^{-3}	—	—	45.39	44.08	87.0	98.0	—	98.0	—	—	−41.61	−53.92
21	Loss weight $\lambda = 1:10$	PANDA	VGG16	10^{-3}	—	—	41.83	38.06	87.0	98.0	—	98.0	—	—	−45.17	−59.94
22	Loss weight $\lambda = 10:1$	PANDA	VGG16	10^{-3}	—	—	44.58	41.52	87.0	98.0	—	98.0	—	—	−42.42	−56.48
Group 5: Ablation Matrix (stain normalization $\diamond$ and skip connections $\times$)																
23	Ablation: No-Macenko $\diamond$	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	40.21	31.57	88.0	99.0	—	99.0	—	—	−47.79	−67.43
24	Ablation: No-Macenko $\diamond$	PanNuke	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	99.36	73.33	—	—	—	—	—	—	—	—
25	Ablation: No-Skips $\times$	TCGA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	94.34	84.12	90.0	98.0	—	98.0	—	—	+4.34	−13.88
26	Ablation: No-Skips $\times$	PANDA	MobileNetV2	10^{-4}	$\checkmark$	$\checkmark$	35.31	28.94	88.0	99.0	—	99.0	—	—	−52.69	−70.06

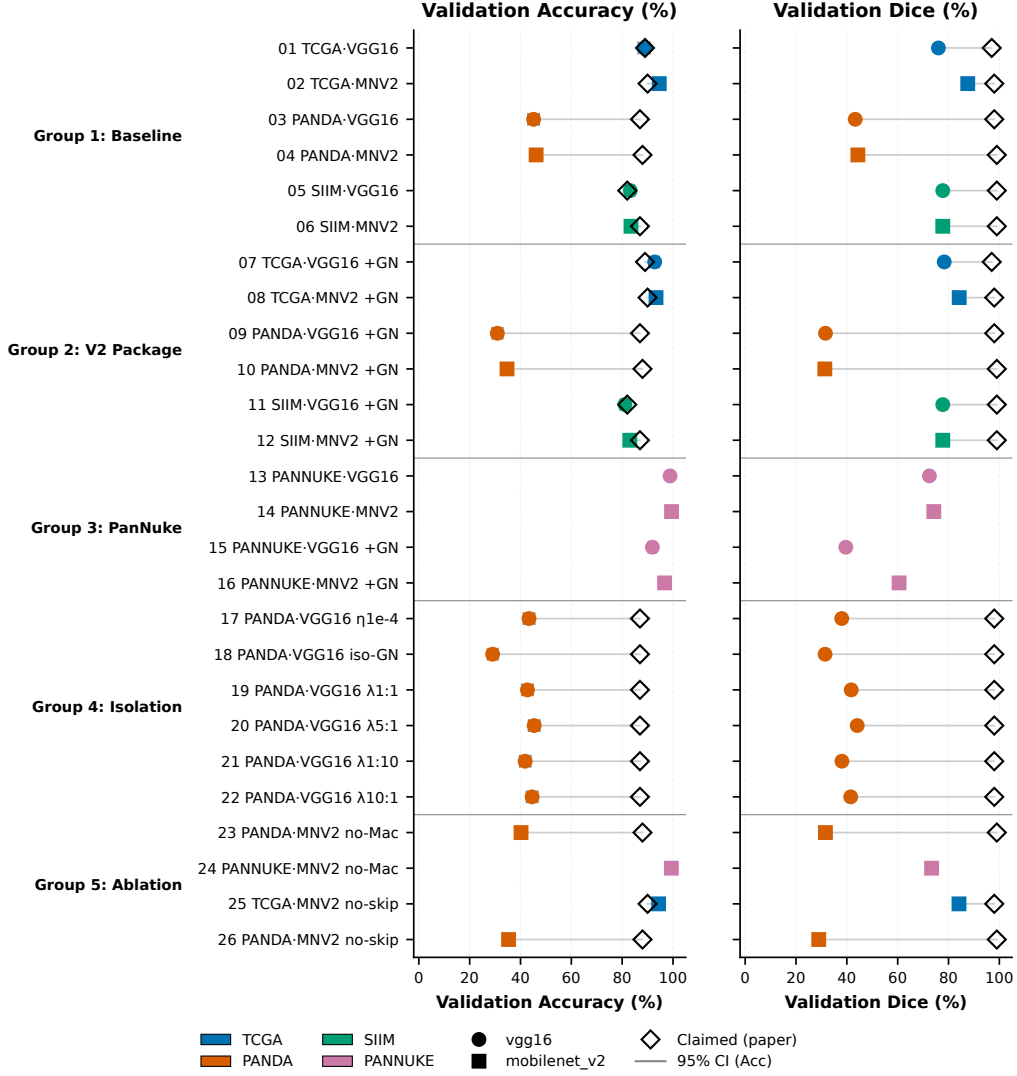


Figure 4: 26-run claimed-vs-measured dot plot. Left panel: validation accuracy (%) with 95% Wilson-CI whiskers (measured, filled markers) and the published claimed accuracy (black open diamond); the tie line between them is the gap that constitutes the paper’s central argument. Right panel: validation Dice (%) with the same layout but no whiskers, as no published confidence intervals exist for the macro-Dice claims. The PANDA rows (Runs 03–04, 09–10, 17–22, 23, 26) show the largest gaps: measured accuracy spans 29.04%–46.15% vs. claimed 87–88%, and Dice spans 28.94%–44.34% vs. claimed 98–99%. All four SIIM rows (Runs 05, 06, 11, 12) measure exactly 77.74% Dice — the all-empty floor. PanNuke rows (Runs 13–16, 24) have no published targets and are shown without claimed diamonds.

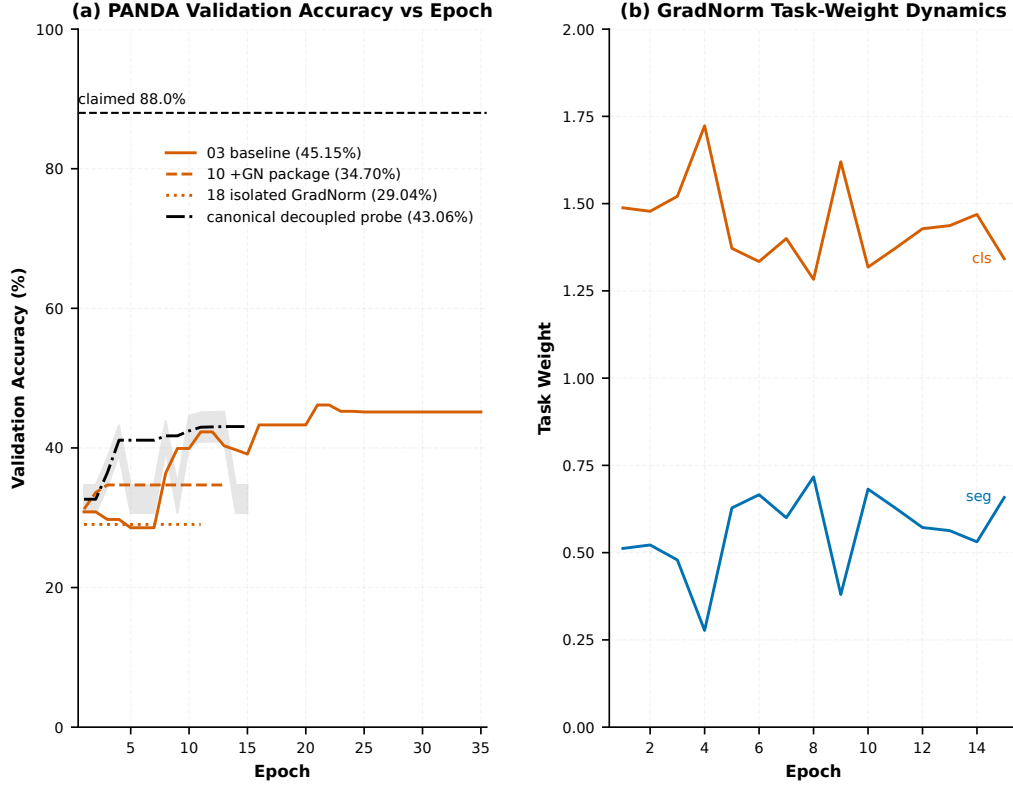


Figure 5: Anatomy of the PANDA collapse. (a) Validation accuracy vs. epoch for three PANDA runs — Run 03 (static 5:1 baseline, 45.15%), Run 10 (v2 GradNorm package, 34.70%), Run 18 (isolated GradNorm at $\eta = 10^{-3}$, 29.04%) — plus the canonical decoupled GRADNORM probe (dash-dot, shaded 95% CI band), which remains stable but plateaus at 43.06% over 15 epochs, 44.94 percentage points below the claimed 88.0% (black dashed reference). (b) GRADNORM task-weight dynamics from the canonical log: the segmentation and classification weights over 15 epochs, showing the dynamic rebalancing that the joint variant turns into destructive updates on the shared encoder.

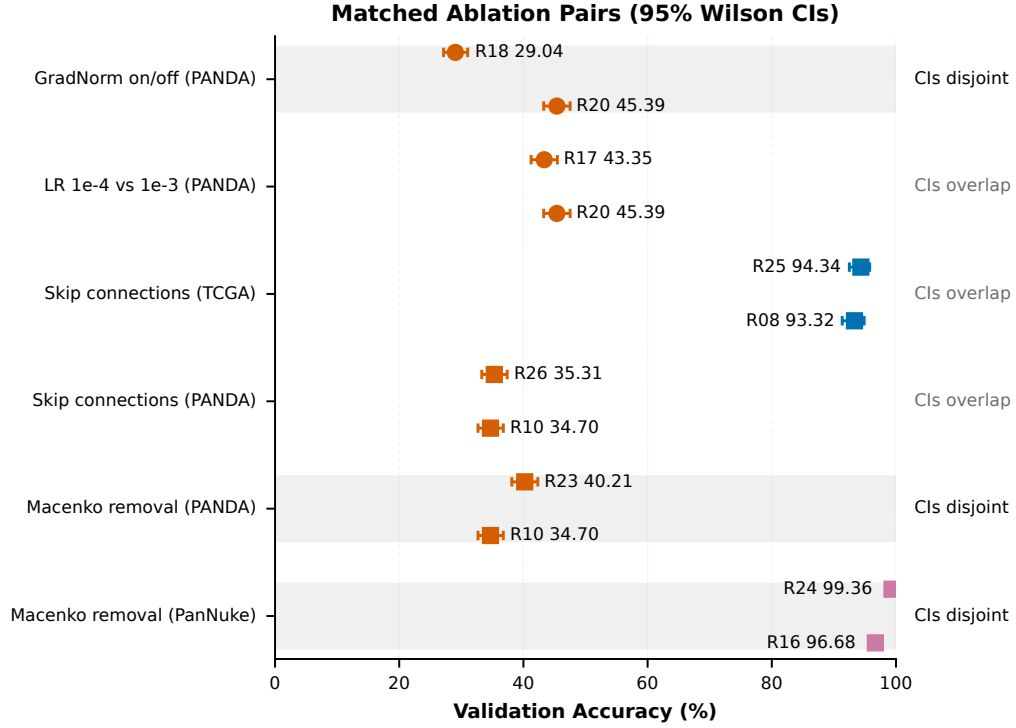


Figure 6: Matched-ablation forest plot of validation accuracy with 95% Wilson CIs for the six strictly matched pairs. Disjoint-CI pairs (shaded): isolated GradNorm (Run 18, 29.04%) vs. static 5:1 (Run 20, 45.39%); no-Macenko PANDA (Run 23, 40.21%) vs. matched (Run 10, 34.70%); no-Macenko PanNuke (Run 24, 99.36%) vs. matched (Run 16, 96.68%). Overlapping-CI pairs: LR 10^{-4} vs. 10^{-3} (Runs 17 vs. 20: 43.35% vs. 45.39%); skip ablation TCGA (Runs 25 vs. 08: 94.34% vs. 93.32%); skip ablation PANDA (Runs 26 vs. 10: 35.31% vs. 34.70%). The right-margin verdict column encodes the CI-overlap test, so the null-result status of the skip ablations and the resolvable Macenko/GradNorm effects are read directly from the plot.

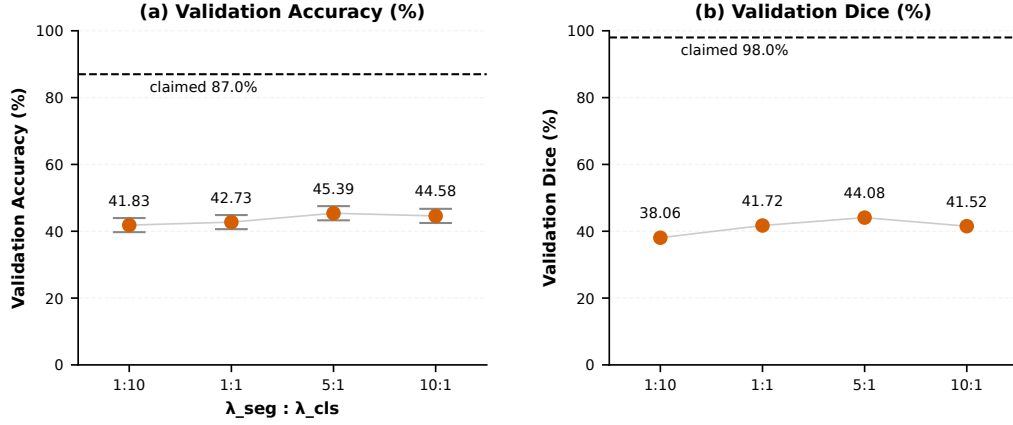


Figure 7: Static loss-weight ratio sweep on PANDA $\times$ VGG16 (raw input, $\eta = 10^{-3}$, no GradNorm). (a) Validation accuracy with 95% Wilson-CI whiskers: $\lambda_{seg} : \lambda_{cls} = 1:10$ (41.83%), 1:1 (42.73%), 5:1 (45.39%), 10:1 (44.58%); the black dashed line marks the claimed 87.0% accuracy. (b) Validation Dice for the same four arms (38.06%, 41.72%, 44.08%, 41.52%) against the claimed 98.0%. The 5:1 ratio is the best static balance on this architecture, yet no ratio approaches the claimed target — the gap is not a loss-weighting artifact.